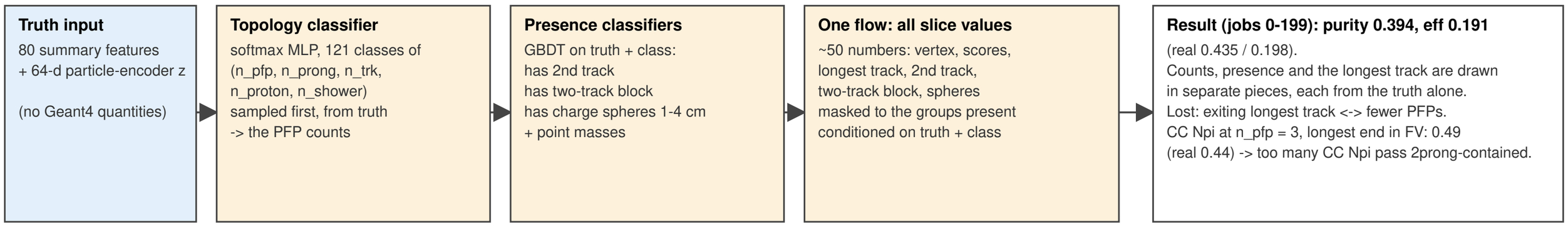
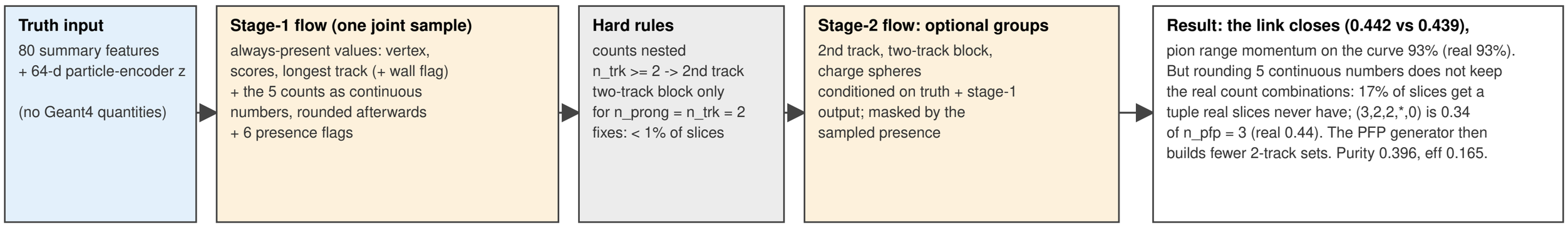
Slice generator: the shipped scheme and the three two-stage variants tried (2026-11-09)

Each row generates one slice record from the truth; the PFP generator (unchanged, run m4_official_m3w0) then builds the PFP set from it. Numbers: CC1p0pi acceptance on test jobs 0-199.

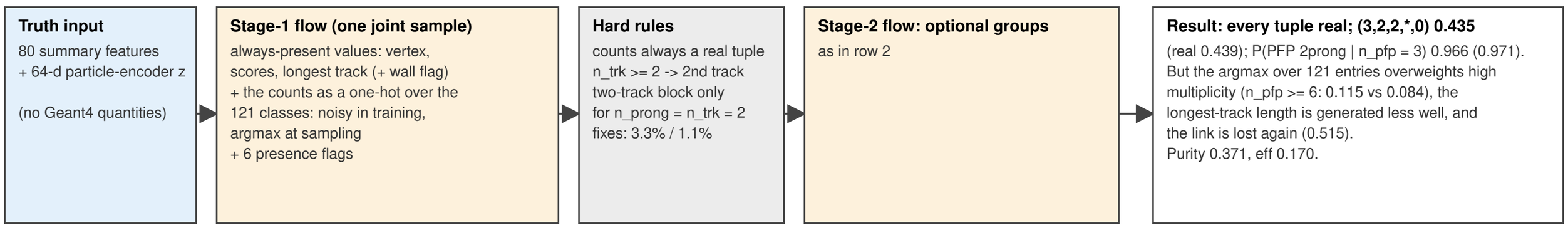
1. Shipped slice generator (run m2_official_m3)



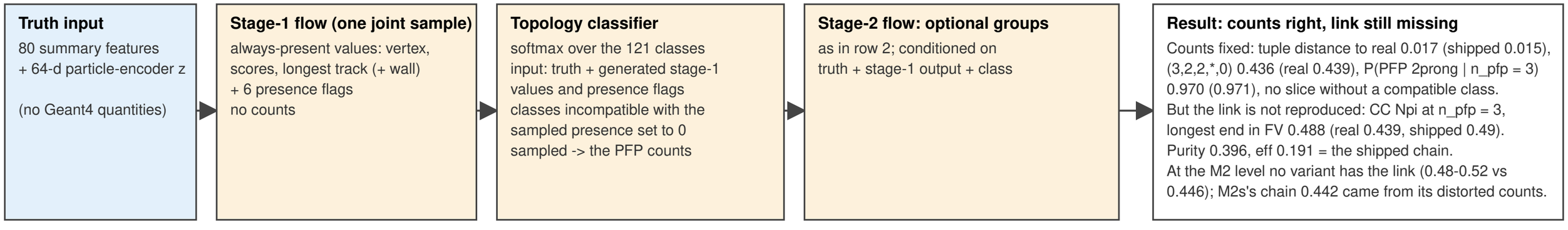
2. Variant A tried (run m2s): two stages, counts as continuous numbers in stage 1



3. Variant B tried (run m2o): two stages, counts as a one-hot class in stage 1



4. Variant C tried (run m2c): two stages, counts from a classifier that reads the generated stage 1



Terms

always-present values: vertex, nu_score, bcfm_score, total_pe, longest-track direction / length / PID / chi2 (every slice has them). optional groups: 2nd track (9 values),
two-track block (reco_t, opening angle, range momenta, 12 end points), charge in 1-4 cm spheres. presence flags: has 2nd track, has two-track block, has sphere 1 / 2 / 3 / 4.
counts: n_pfp, n_prong, n_trk, n_proton, n_shower; 121 real combinations + other. the link: in real data a slice whose longest track leaves the TPC tends to have fewer PFPs,
so CC Npi events with an exiting muon fall into n_pfp = 3 (the 2prong bin) more often. Colours: blue truth input, orange learned model, grey deterministic rule, white result.
Models: particle encoder (code m3), slice generator (code m2), PFP generator (code m4).